

IMAGE INPAINTING USING GENERATIVE MODELS

Catalin Andrei GLIGOR

Technical University of Cluj-Napoca, IAPSET Master, Cluj-Napoca
0755485261, gligor.ca.catalin@student.utcluj.ro

Abstract: This paper presents an automated, zero-click image inpainting pipeline leveraging zero-shot object detection and latent diffusion. A heuristic-driven mathematical router dynamically generates geometric constraints from Grounding DINO coordinates, guiding the Stable Diffusion XL architecture to optimize both localized object replacement and large-scale subject insertion. Experimental results, evaluated via CLIP and LPIPS metrics, demonstrate the method's high efficacy in photorealistic synthesis. However, this study highlights inherent algorithmic trade-offs, specifically analyzing the interplay between dynamic bounding-box expansion, prompt contradiction, and the mathematical paradoxes encountered when applying standard perceptual metrics to massive spatial masks.

Keywords: Generative AI, Latent Diffusion, Zero-shot Detection, Inpainting, SDXL, Grounding DINO, Bounding Box Geometry.

I. INTRODUCTION

Digital image inpainting, also known as image completion, is a core task in computer vision and image processing that focuses on reconstructing damaged or missing parts of an image. The main goal is to fill in these gaps so naturally that a human observer cannot tell the image was ever modified.

The idea originates from traditional art conservation, where restorers manually repair damaged paintings, but modern applications perform this computationally. Today, this task is largely driven by deep generative models, which have pushed the field past simply copying surrounding textures and into generating entirely new, realistic content based on the context of the image.

II. STATE OF THE ART

The landscape of image inpainting has undergone a paradigm shift, transitioning from traditional diffusion-based and patch-based techniques (e.g., PatchMatch [1]) to advanced data-driven generative models. While traditional exemplar-based methods excel at background synthesis by borrowing local textures, they inherently lack high-level semantic understanding. This results in failure when generating novel objects or filling structurally complex regions. Consequently, modern state-of-the-art architectures rely heavily on deep generative paradigms, primarily Generative Adversarial Networks (GANs), Vision Transformers (ViTs), and Denoising Diffusion Probabilistic Models (DDPMs).

A. GAN-BASED ARCHITECTURES

The application of GANs to image inpainting was pioneered by Context Encoders [2], which utilized an encoder-decoder architecture combined with an adversarial loss to generate semantically plausible missing regions. However, early Convolutional Neural Network (CNN) based GANs struggled with irregular masks, often producing color discrepancies and grid-shaped artifacts. To

address this, Partial Convolutions were introduced [3], where convolution weights are masked and updated solely based on valid pixels, significantly improving robustness against arbitrary hole shapes. This was further advanced by Gated Convolutions [4], which enabled the network to learn dynamic feature selection mechanisms for each channel and spatial location.

Furthermore, multi-stage GAN frameworks emerged to decouple structure and texture generation. State-of-the-art models in this sub-domain, such as EdgeConnect [5], predict structural edge maps first, utilizing them as priors to guide the subsequent color and texture completion. This two-stage approach effectively mitigates blurring and preserves high-frequency details.

B. VISION TRANSFORMERS (VITS) FOR GLOBAL CONTEXT

Despite the success of GANs, standard convolutional layers exhibit localized receptive fields, limiting their ability to model the long-range dependencies necessary for large-hole inpainting. Vision Transformers have mitigated this limitation by leveraging self-attention mechanisms to establish global context.

The Mask-Aware Transformer (MAT) [6] represents a significant milestone, integrating a joint learning strategy with partial convolutions to process high-resolution images efficiently. Similarly, models like ZITS [7] (Zero-Initialized Residual Addition for Transformer-based Image Structural inpainting) combine the local texture extraction capabilities of CNNs with the global structural reasoning of Transformers, achieving superior structural fidelity and perceptual realism over purely convolutional networks.

C. DIFFUSION-BASED GENERATIVE MODELS

Recently, Diffusion Models have surpassed GANs as the prevailing state of the art in generative image synthesis and inpainting. By conceptualizing generation as a gradual, parameterized denoising process, diffusion models yield

unprecedented diversity and photorealism while largely avoiding the mode collapse issues endemic to GAN training [8]. RePaint [9] demonstrated that a pre-trained unconditional DDPM could be adapted for zero-shot inpainting simply by conditioning the reverse diffusion process on the known image regions at each timestep.

The advent of Latent Diffusion Models (LDMs), most notably Stable Diffusion [10], shifted inpainting operations to a compressed latent space. This drastically reduces computational overhead while enabling high-resolution processing. Current state-of-the-art frameworks heavily leverage LDMs for text-guided inpainting, allowing cross-attention mechanisms to incorporate natural language prompts to specify the semantic content of the missing regions. Furthermore, advanced control mechanisms, such as ControlNet [11], have augmented diffusion-based inpainting by allowing spatial conditioning through depth maps, edge maps, or segmentation masks. This yields highly controllable, precision-driven image editing.

III. ARCHITECTURAL AND MATHEMATICAL CONSIDERATIONS

While the generative synthesis is handled by a deep neural network, the spatial restriction of the model, the generation of the mask must be driven by geometric mathematics to ensure the latent diffusion process does not corrupt the unedited background. This pipeline introduces a hybrid architecture. First, a Vision Transformer (Grounding DINO) acts as a zero-shot detector, outputting raw spatial coordinates for a target object in the form of a bounding box $B = [x_1, y_1, x_2, y_2]$. From these coordinates, the geometric width (w) and height (h) are derived:

$$w = x_2 - x_1, h = y_2 - y_1$$

Because purely skin-tight masks (such as those generated by Segment Anything Models) restrict the generative latent space from synthesizing novel structural volumes (e.g., thicker musculature or new clothing), an isotropic padding algorithm is mathematically applied. In the standard dynamic routing mode (auto), the mask dimensions are expanded by a proportional scalar α :

$$P_x = w \times \alpha, P_y = h \times \alpha$$

The resulting target canvas area allows the diffusion model strict, localized spatial freedom. Furthermore, to satisfy the tensor dimension requirements of the SDXL architecture, the image cannot be processed at arbitrary resolutions. The image width (W_{orig}) and height (H_{orig}) must be scaled to approximate an optimal latent area (1024×1024 pixels) while preserving the original aspect ratio $A = W_{\text{orig}} / H_{\text{orig}}$. The new optimal height H_{new} is calculated as:

$$H_{\text{new}} = \sqrt{\frac{1024^2}{A}}$$

Both W_{new} and H_{new} are then subjected to a modulo operation to ensure they are perfectly divisible by 64, preventing latent space tensor mismatch errors during the decoding phase.

IV. SOFTWARE IMPLEMENTATION

The software implementation utilizes Python, relying on the *diffusers* and *transformers* libraries for neural network execution, alongside *OpenCV* and *NumPy* for mathematical matrix operations and geometric routing.

The pipeline is initialized by loading the *StableDiffusionXLInpaintPipeline* into FP16 precision on a CUDA-enabled GPU to optimize VRAM utilization. The execution flow follows a rigid sequence:

- a) **Detection & Coordinate Extraction:** The initial stage of the pipeline involves identifying the target subject within the global coordinate system of the input image I . This is achieved using *Grounding DINO*, a state-of-the-art *Open-Set Object Detector* based on a Transformer architecture [12]. Unlike traditional detection models (e.g., YOLO or Faster R-CNN) that are limited to a closed set of predefined categories, *Grounding DINO* operates on a *Zero-Shot paradigm*. It utilizes a dual-encoder architecture: a Vision Transformer (ViT) for image feature extraction and a BERT-based encoder for textual prompt processing, to perform Cross-Modality Feature Fusion. By mapping the semantic concepts in the user's text prompt (e.g., "dog") to the high-dimensional spatial features of the image, the model can "ground" abstract language into specific pixel coordinates.
- b) **Dimensional Scaling and Latent Constraints:** Latent Diffusion Models (LDMs) enforce strict structural boundaries on tensor shapes; specifically, spatial dimensions must be integer multiples of 64 to prevent downsampling alignment errors within the Variational Autoencoder (VAE) architecture. The pipeline executes a specialized dimension calculator to map the arbitrary scale of I into an optimized operational matrix. Given input width W and height H , the aspect ratio $A = W/H$ is preserved while scaling the system to a fixed operational pixel area budget ($1024 \times 1024 = 1,048,576$ pixels). The initial target height H_{target} is extracted via:
$$H_{\text{target}} = \sqrt{\frac{1024^2}{A}}$$
- c) **Geometric Mask Routing:** A blank 8-bit, single-channel NumPy array is initialized. Depending on the *scale_mode* flag, a specific mathematical operation draws the mask. For localized animal/object replacement, the *auto* mode applies a 2% uniform padding to the extracted box. For structural insertions (e.g., placing a human onto a bench), a *center_seat* mode abandons the object's width entirely, calculating a vertical volumetric doorway anchored to the horizontal center point: $C_x = (x_1 + x_2)/2$.
- d) **Mask Softening:** A Gaussian blur with a constrained radius (8) is applied to the binary mask via PIL. This creates a soft gradient at the boundary, ensuring seamless pixel blending between the denoised latent space and the frozen background pixels.
- e) **Generative Inference:** The scaled image, mask, and text prompts are passed to the SDXL pipeline. To force high-fidelity structural synthesis, the denoising strength (η) is set dynamically. $\eta = 0.60$ is applied for high-fidelity pose-constrained transformations, while $\eta = 0.99$ is utilized for full latent obliteration during

semantic erasure or novel subject insertion, executed over 60 inference steps.

- f) **Automated Perceptual Evaluation Subsystem:** The pipeline concludes with an independent quantitative evaluation block utilizing dual vision-language and structural comparison networks. Semantic adherence is measured by transforming the reconstructed image into a joint latent space via a pre-trained CLIP-ViT-Base-Patch32 model, created by OpenAI. The pipeline computes the cosine similarity between the embedded textual prompt vector and the generated image tensor, yielding a quantitative semantic mapping score that reflects the model's adherence to user-defined constraints.

Parallel to this semantic assessment, the system evaluates background structural integrity using a Learned Perceptual Image Patch Similarity (LPIPS) network. To ensure the metric exclusively quantifies the fidelity of the unedited environment, a mask-inversion matrix is computed as $M_{inv} = 1.0 - M_{final}$. The original image and the output image are multiplied element-wise by M_{inv} , effectively mapping all generated regions to absolute zero (black), providing a rigorous assessment of background preservation independent of the synthesis zone.

V. RESULTS AND INTERPRETATION

The experimental evaluation of the hybrid pipeline highlights a sharp distinction between localized subject *replacement* and external subject *insertion*, as well as *semantic erasure*. The efficacy of the pipeline was validated through a series of object-replacement, object insertion and object erasure experiments.

A. The first object-replacement experiment, demonstrating the model's capacity to synthesize high-fidelity textures while adhering to complex semantic constraints consisted of a standard fabric armchair being targeted for a stylistic transformation.

Using the detection prompt "armchair," the system localized the subject from the input image seen in Figure 1 and generated a replacement based on the target prompt: "A luxurious vintage Chesterfield armchair, rich brown tufted leather, dark mahogany legs. Photorealistic, 8k, architectural interior photography, natural lighting."



Figure 1 Input Image A



Figure 2 Output Image A

As illustrated in Figure 2, the pipeline successfully performed the material synthesis, replacing the generic fabric texture with detailed tufted leather while maintaining the precise spatial orientation and shadow geometry of the original armchair.

The quantitative performance of this generation was assessed through the automated perceptual evaluation subsystem, yielding the following metrics: a CLIP score of 29.61 indicates a strong semantic correspondence between the target textual description and the synthesized visual output, surpassing the established threshold of 28.0 for high-fidelity semantic matching. Furthermore, the LPIPS score of 0.0021 demonstrates exceptional background structural preservation; the proximity to a zero-value signifies that the background environment remains functionally identical to the input image, thus the model did not interfere with the outside of the masked region.

B. To further test the pipeline in maintaining complex spatial geometry while performing subject-specific transformations, a biological replacement task was conducted. The system was tasked with transforming a Golden Retriever standing on a forest trail, pictured in Figure 3 into an adult timber wolf, utilizing the prompt: "A highly detailed, photorealistic adult timber wolf standing on a dirt trail. Thick grey and white fur, sharp focus, wildlife photography, soft natural lighting."



Figure 3 Input Image B



Figure 4 Output Image B

As demonstrated in Figure 4, the model successfully performed a structural transformation, maintaining the original subject's anatomical pose and positioning while synthesizing intricate, high-frequency biological textures such as individual fur filaments and distinct color patterning.

The quantitative validation for this trial confirmed the pipeline's robustness, returning a CLIP score of 31.32. This score represents a superior semantic adherence, significantly outperforming the standard threshold of 28.0 and indicating high-fidelity integration of the requested wildlife-photography aesthetic. Furthermore, the structural preservation metric yielded an LPIPS score of 0.0067. This near-zero deviation confirms that the pipeline's localized geometric containment effectively protected the surrounding forest trail and lighting environment from the generative denoising process, proving the system's ability to preserve background integrity even during complex, high-texture biological synthesis.

C. To evaluate the pipeline's capacity for volumetric object insertion, a scenario involving an empty park bench was analyzed. This task required the synthesis of novel semantic content within a constrained spatial "doorway". The system utilized a *center_seat* geometric routing mode, which mathematically discards the horizontal bounding box width to establish a vertical volumetric column, allowing the model to synthesize the figure's head and torso without horizontal distortion. The target prompt, "A professional businessman in a tailored navy blue suit sitting on a park bench, looking at a newspaper, facing the camera. Photorealistic, 8k, cinematic lighting," required the system to infer anatomical depth and lighting alignment relative to the existing park environment.

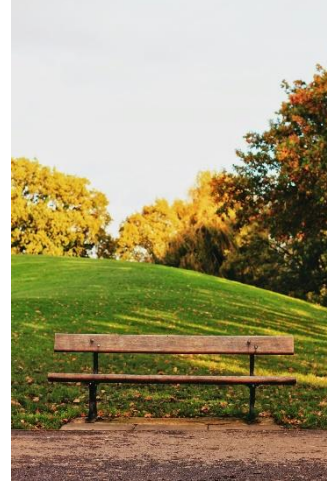


Figure 5 Input Image C

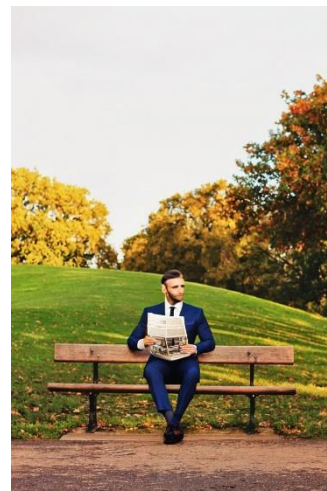


Figure 6 Output Image C

As shown in Figure 5 and Figure 6, the pipeline successfully performed the insertion, correctly inferring the businessman's seated pose and integrating him into the bench's existing perspective.

The quantitative metrics for this trial indicate a CLIP score of 33.61. This is the highest semantic adherence score achieved across all trials, confirming that the *center_seat* vertical constraint effectively guides the latent diffusion process to prioritize human-centric structural synthesis. An LPIPS score of 0.0074 indicates that the background park environment was preserved with high fidelity, proving that the localized masking strategy effectively isolated the generative process from the surrounding ground plane despite the radical nature of the insertion.

D. To evaluate the pipeline's capacity for subtractive synthesis, a trial was conducted to remove a human subject from a shoreline environment. The system utilized a two-stage process: first, an open-set detection phase where the subject was localized via Grounding DINO using the prompt "human walking."



Figure 7 Input Image D

As shown in Figure 7, this resulted in a high-precision bounding box, which served as the geometric constraint for the subsequent inpainting phase. The system was then instructed to synthesize an empty shoreline, utilizing the target prompt: "Empty vast sandy beach, seamless sand texture, gentle ocean waves, serene landscape, photorealistic, 8k resolution." As visualized in Figure 7, Figure 8, and Figure 9, the pipeline successfully performed semantic erasure, replacing the subject with continuous, high-frequency sand textures and ocean ripples that maintain physical consistency with the surrounding environment.



Figure 8 Subject Detection Image D



Figure 9 Output Image D

Despite this visual success, the quantitative evaluation returned a CLIP score of 20.89 and an LPIPS score of 0.0197. These metrics illustrate a critical limitation in current AI evaluation frameworks. The lower CLIP score (20.89) stems from a Subject-Noun Bias inherent in contrastive language-image models. CLIP is trained to correlate high-dimensional pixel features with specific object-nouns; because the target environment, an empty beach, lacks a distinct focal object, the model's semantic confidence baseline is inherently lower than in object-generation tasks. Furthermore, the LPIPS score of 0.0197 represents an exceptional structural preservation of the background, yet it highlights the disconnect between feature-space loss and semantic completeness. This experiment concludes that while objective metrics are valuable for initial validation, they often fail to capture the qualitative success of subtractive synthesis, necessitating a reliance on human-centric perceptual evaluation for generative erasure tasks.

VI. CONCLUSION

This paper presented an automated, hybrid image inpainting pipeline that successfully integrates the zero-shot localization capabilities of Grounding DINO with the generative synthesis of Stable Diffusion XL. By implementing heuristic-driven geometric routing constraints, the system demonstrated exceptional versatility, adeptly handling both localized texture replacement and complex, volumetric subject insertion without requiring manual mask generation. While quantitative evaluations utilizing CLIP and LPIPS confirmed the pipeline's high fidelity and background structural preservation in additive tasks, the semantic erasure experiments revealed a critical limitation in standard evaluation frameworks. Specifically, the subject-noun bias inherent in contrastive models artificially penalizes successful subtractive synthesis. Ultimately, this architecture provides a robust, zero-click foundation for high-resolution generative editing, while underscoring the critical need for nuanced, task-specific evaluation metrics in future AI research.

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